

AAE 50700: Principles of Dynamics
Problem Set 5 Report

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(5.1) Orbital Dynamics with J_2 and Atmospheric Drag

(a): Given

Consider a spacecraft of mass m orbiting Earth perturbed by J_2 and atmosphere drag. Let the inertial frame be $\mathcal{N} : \{\hat{e}_x, \hat{e}_y, \hat{e}_z\}$ and write its Cartesian position as $\mathbf{r} = x\hat{e}_x + y\hat{e}_y + z\hat{e}_z, r = \|\mathbf{r}\|_2$. Expressing the potential energy for Earth gravity under J_2 in the Cartesian coordinates, we have

$$V(x, y, z) = -\frac{\mu m}{r} + \frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2), \quad (1)$$

where μ is Earth's gravitational parameter and $r_o = R_{\text{earth}}$ is its equatorial radius.

We use the same model as considered in PS1 to model the drag, which assumes an inertially stationary, spherically symmetric atmosphere, given by

$$\mathbf{F}_D = m\mathbf{a}_D, \quad \mathbf{a}_D = -\frac{1}{2}C_D \frac{A}{m} \rho(r) v^2 \hat{\mathbf{v}}, \quad v := \|\mathbf{v}\|_2, \quad \hat{\mathbf{v}} := \frac{\mathbf{v}}{v}, \quad (2)$$

where $\mathbf{v} = \dot{\mathbf{r}}$, C_D is the drag coefficient, A is the reference area, and

$$\rho(r) \approx \rho_0 \exp \left[-\frac{r - 6978 \text{ km}}{H} \right]. \quad (3)$$

(a.1): Find

From the Lagrangian using the Cartesian coordinates as generalized coordinates, $\mathbf{q} = [x, y, z]^T$.

Analysis/Conclusions:

First recall that the kinetic energy T of an object in motion can be expressed by half the mass times the square of the velocity:

$$T(\mathbf{v}) = \frac{m}{2} \mathbf{v}^T \mathbf{v} = \frac{mv^2}{2} \quad (4)$$

Then, recall the Lagrangian L is the kinetic energy minus the potential energy $L = T - V$ (V given by the problem definition):

$$L(\mathbf{r}, \mathbf{v}) = \frac{mv^2}{2} + \frac{\mu m}{r} - \frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \quad (5)$$

(a.2): Find

Derive the generalized force for Eq. (2) associated with $\mathbf{q} = [x, y, z]^T$.

Analysis/Conclusions:

It is known that virtual work δw is translational force $\mathbf{F}$ dotted with the Jacobian of position with respect to the generalized coordinates multiplied by the displacement of generalized coordinates. This also equals generalized force $\mathbf{Q}$ dotted with the same displacement:

$$\delta w = \mathbf{F}^T \frac{\partial \mathbf{r}}{\partial \mathbf{q}} \delta \mathbf{q} = \mathbf{Q}^T \delta \mathbf{q}. \quad (6)$$

Since in this case the generalized coordinates are the Cartesian position, $\mathbf{q} = \mathbf{r}$, the Jacobian is the identity matrix, and the generalized force is simply the applied force:

$$\delta w = \mathbf{F}^T \frac{\partial \mathbf{r}}{\partial \mathbf{r}} \delta \mathbf{r} = \mathbf{F}^T I \delta \mathbf{r} = \mathbf{F}^T \delta \mathbf{r} \Rightarrow \mathbf{Q} = \mathbf{F}. \quad (7)$$

So, from Eq. (2) there is only one force, which is the force of drag $\mathbf{F}_D$. This means the generalized force from that equation associated with $\mathbf{q}$ is $\mathbf{Q} = \mathbf{F}_D$. The mass terms cancel and simplify to get generalized force:

$$\mathbf{Q} = \mathbf{F}_D = m \mathbf{a}_D = m \left(-\frac{1}{2} C_D \frac{A}{m} \rho(r) v^2 \hat{\mathbf{v}} \right) = -\frac{1}{2} C_D A \rho(r) v^2 \hat{\mathbf{v}} \quad (8)$$

(a.3): Find

Derive the Lagrange's equations and wrtie the result as the Cartesian vector EOM

$$\ddot{\mathbf{r}} = \mathbf{a}_{2B} + \mathbf{a}_{J2} + \mathbf{a}_D \quad (9)$$

We see here that this system has three degrees a freedom with no redundancy in coordinates, so $\mathbf{q}$ is a minimal coordinate set, and the system has no constraints on movement. Since there is also a non-conservative force (drag), the general form of Lagrange's equations with non-conservative general force may be used:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial L}{\partial \mathbf{q}} = \mathbf{Q}_{nc} \quad (10)$$

And since, $\mathbf{q} = \mathbf{r}$ and $\mathbf{F}_D$ is the only non-conservative general force:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\mathbf{v}}} \right) - \frac{\partial L}{\partial \mathbf{r}} = \mathbf{F}_D \quad (11)$$

Then we compute the derivatives in Eq. (11) individually.

So first to solve $\frac{\partial L}{\partial \mathbf{v}}$:

$$\begin{aligned}\frac{\partial L}{\partial \mathbf{v}} &= \frac{\partial}{\partial \mathbf{v}} \left(\frac{mv^2}{2} + \frac{\mu m}{r} - \frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\ &= \frac{m}{2} \frac{\partial v^2}{\partial \mathbf{v}} + \frac{\partial}{\partial \mathbf{v}} \left(\frac{\mu m}{r} \right) - \frac{\partial}{\partial \mathbf{v}} \left(\frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\ &= \frac{m}{2} 2\mathbf{v}^T + 0 - 0 = m\mathbf{v}^T\end{aligned}\tag{12}$$

Then to solve $\frac{d}{dt} \left(\frac{\partial L}{\partial \mathbf{v}} \right)$:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \mathbf{v}} \right) = m \frac{d\mathbf{v}^T}{dt} = m\mathbf{a}^T\tag{13}$$

Then solve $-\frac{\partial L}{\partial \mathbf{r}}$:

$$\begin{aligned}-\frac{\partial L}{\partial \mathbf{r}} &= -\frac{\partial}{\partial \mathbf{r}} \left(\frac{mv^2}{2} + \frac{\mu m}{r} - \frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\ &= -\frac{\partial}{\partial \mathbf{r}} \left(\frac{mv^2}{2} \right) - \mu m \frac{\partial r^{-1}}{\partial \mathbf{r}} + \frac{\partial}{\partial \mathbf{r}} \left(\frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\ &= 0 - \mu m \frac{\partial r^{-1}}{\partial \mathbf{r}} + \frac{\partial}{\partial \mathbf{r}} \left(\frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right)\end{aligned}\tag{14}$$

Then for readability, we do the remaining component derivatives individually:

$$\frac{\partial r^{-1}}{\partial \mathbf{r}} = \frac{\partial r^{-1}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} = -r^{-2} \hat{\mathbf{r}}^T\tag{15}$$

And the other component:

$$\begin{aligned}
&= \frac{\partial}{\partial \mathbf{r}} \left(\frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(\frac{\partial}{\partial \mathbf{r}} (3r^{-5} z^2) - \frac{\partial r^{-3}}{\partial \mathbf{r}} \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left(\frac{\partial r^{-5}}{\partial \mathbf{r}} z^2 + r^{-5} \frac{\partial z^2}{\partial \mathbf{r}} \right) - \frac{\partial r^{-3}}{\partial \mathbf{r}} \frac{\partial r}{\partial \mathbf{r}} \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left(\frac{\partial r^{-5}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} z^2 + r^{-5} \frac{\partial z^2}{\partial z} \frac{\partial z}{\partial \mathbf{r}} \right) - \frac{\partial r^{-3}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left(\frac{\partial r^{-5}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} z^2 + r^{-5} \frac{\partial z^2}{\partial z} \hat{\mathbf{e}}_z^T \frac{\partial \mathbf{r}}{\partial \mathbf{r}} \right) - \frac{\partial r^{-3}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left((-5r^{-6})(\hat{\mathbf{r}}^T) z^2 + 2r^{-5} z \hat{\mathbf{e}}_z^T I \right) - (-3r^{-4} \hat{\mathbf{r}}^T) \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left((-5r^{-6})(r^{-1} \mathbf{r}^T) z^2 + 2r^{-5} z \hat{\mathbf{e}}_z^T I \right) - (-3r^{-4})(r^{-1} \mathbf{r}^T) \right) \\
&= \frac{3}{2} \mu m J_2 r_o^2 \left(-5r^{-7} z^2 \mathbf{r}^T + 2r^{-5} z \hat{\mathbf{e}}_z^T + r^{-5} \mathbf{r}^T \right) \\
&= \frac{3}{2} \mu m J_2 r_o^2 r^{-5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right) \\
&= \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right)
\end{aligned} \tag{16}$$

So then that gives:

$$\begin{aligned}
-\frac{\partial L}{\partial \mathbf{r}} &= -\mu m \left(-r^{-2} \hat{\mathbf{r}}^T \right) + \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right) \\
&= \frac{\mu m}{r^3} \mathbf{r}^T + \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right)
\end{aligned} \tag{17}$$

Which then plugging the above terms back into Eq. (11) gives:

$$m\mathbf{a}^T + \frac{\mu m}{r^3} \mathbf{r}^T + \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right) = -\frac{1}{2} C_D A \rho(r) v^2 \hat{\mathbf{v}}^T \tag{18}$$

Which then solved for acceleration $\mathbf{a} = \ddot{\mathbf{r}}$:

$$\begin{aligned}
m\mathbf{a} &= -\frac{\mu m}{r^3} \mathbf{r} - \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z \hat{\mathbf{e}}_z \right) - \frac{1}{2} C_D A \rho(r) v^2 \hat{\mathbf{v}} \\
\mathbf{a} &= -\frac{\mu m}{r^3 m} \mathbf{r} - \frac{3\mu m J_2 r_o^2}{2r^5 m} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z \hat{\mathbf{e}}_z \right) - \frac{1}{2m} C_D A \rho(r) v^2 \hat{\mathbf{v}} \\
\mathbf{a} &= -\frac{\mu}{r^3} \mathbf{r} - \frac{3\mu J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z \hat{\mathbf{e}}_z \right) - \frac{1}{2m} C_D A \rho(r) v^2 \hat{\mathbf{v}}
\end{aligned} \tag{19}$$

Which means the Cartesian vector EOM are:

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r} - \frac{3\mu J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z\hat{\mathbf{e}}_z \right) - \frac{1}{2m}C_D A \rho(r) v^2 \hat{\mathbf{v}} \quad (20)$$

Where:

$$\begin{aligned} \mathbf{a}_{2B} &= -\frac{\mu}{r^3}\mathbf{r} \\ \mathbf{a}_{J2} &= -\frac{3\mu J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z\hat{\mathbf{e}}_z \right) \\ \mathbf{a}_D &= -\frac{1}{2m}C_D A \rho(r) v^2 \hat{\mathbf{v}} \end{aligned} \quad (21)$$

Since these terms come from the two-body problem, J_2 perturbation, and drag acceleration components of the system respectively.